# Machine Learning for Customer Segmentation Using K-Means Clustering

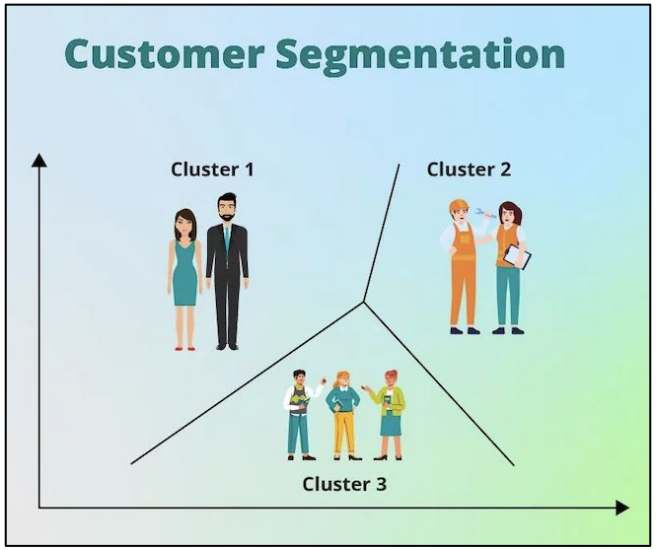
*How K-Means can help businesses find different types of customers*

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# Agenda

1. **Why customer segmentation matters**
2. **K-Means and related research**
3. **Dataset and experiment setup**
4. **Results and customer group interpretation**
5. **Limitations and key takeaways**

# Why Customer Segmentation Matters




- Customers have different needs, budgets, and spending habits.
- Segmentation groups customers with similar characteristics.
- Businesses can use these groups and patterns to inform marketing, product recommendations, and customer engagement.

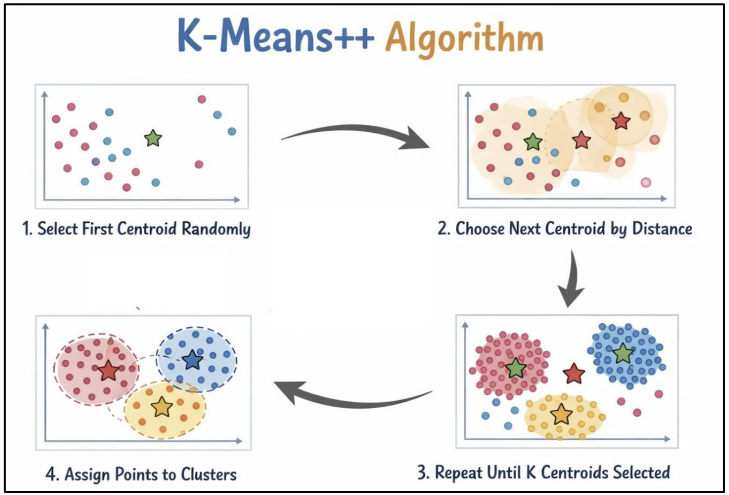
**Say:** Segmentation helps businesses understand who their customers are rather than relying on one strategy for everyone. Segmentation divides customer base into smaller, manageable groups based on behavior, demographics, or interests. **Example:** Amazon studies what you browse, buy, and save so it can predict what you like. This lets it show personalized recommendations that boost how much customers spend.

# What Is K-Means Clustering?

- K-Means is an unsupervised machine learning algorithm.
- It divides observations into K clusters.
- Each cluster is represented by a centroid.
- Observations are assigned to the nearest centroid based on distance.
- The goal is to reduce the variation within each cluster.

**Say:** Explain that K is the number of clusters you request. For example, setting K=3 asks the algorithm to create three groups. Since the dataset does not provide known customer-group labels, this is an unsupervised learning task.

# How K-Means Works




1. Choose the number of clusters, K.
2. Initialize the centroids.
3. Assign each customer to the nearest centroid.
4. Recalculate each centroid using its assigned customers.
5. Repeat assignment and updating until the clusters stabilize.

K-Means begins by choosing the number of clusters and initializing their centroids. Each customer is assigned to the nearest centroid (center; average position), and the centroids are recalculated based on the assigned customers. This process repeats until the clusters stabilize. The final result depends on the features we select and the number of clusters we choose.

# Related Research & Algorithm Variants

- Ikotun et al. (2023): comprehensive review of K-Means algorithms, variants, and advances.
- K-Means remains a widely studied clustering approach, but algorithm variants address different computational and clustering challenges.
- Customer segmentation is one application of clustering.
- My experiment investigates how cluster selection affects the interpretation of customer data.

My research included Ikotun's review of K-Means algorithms, their variants, and recent improvements. It helped explain why clustering methods remain an active area of study. For my experiment, I looked at how changing the number of clusters shifts the customer groups and how meaningful those groups are to interpret.

# Dataset & Selected Features

- **Number of Records:** 200 customers
- **Feature 1:** Annual Income (k$)
- **Feature 2:** Spending Score (1–100)
- **Objective:** Identify customer groups based on income and spending score.

Annual income is expressed in thousands of dollars, while spending score is a score on a 1–100 scale. These are the two dimensions used to form the clusters.

# Setup & Preprocessing

```
results = []
models = {}
labels_by_k = {}

for k in K_VALUES:
    if k >= len(X):
        warnings.warn(f"Skipping K={k}: K must be smaller than the number of observations.")
        continue

    model = KMeans(
        n_clusters=k,
        random_state=RANDOM_STATE,
        n_init=10
    )
    labels = model.fit_predict(X_scaled)
    unique_labels = np.unique(labels)

    # Silhouette is defined only when there are at least 2 and fewer than n_samples labels.
    if 2 <= len(unique_labels) < len(X):
        sil = silhouette_score(X_scaled, labels)
    else:
        sil = np.nan
        warnings.warn(f"Silhouette score is undefined for K={k} in this run.")

    results.append({
        "K": k,
        "Inertia": float(model.inertia_),
        "Silhouette Score": float(sil) if not np.isnan(sil) else np.nan,
        "Clusters Found": int(len(unique_labels))
    })
    models[k] = model
    labels_by_k[k] = labels

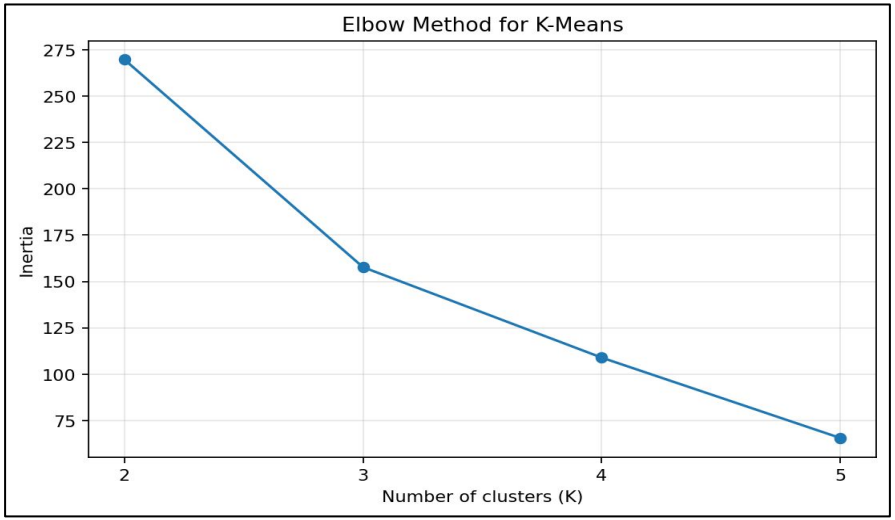
results_df = pd.DataFrame(results)
print("Evaluation results from this run:")
display(results_df.round(4))

results_df.to_csv(OUTPUT_DIR / "kmeans_evaluation_results.csv", index=False)
print("Saved:", OUTPUT_DIR / "kmeans_evaluation_results.csv")
```

- Selected the two numerical features.
- Standardized the features before clustering.
- Tested K=2,3,4,5.
- Used Python and scikit-learn K-Means implementation.
- Set random_state=42 and n_init=10.
- Retained all 200 records.

I used Python and scikit-learn to run K-Means on 200 customer records. I first scaled annual income and spending score so their different number ranges wouldn't affect the distance calculations. Then I tried four cluster amounts—two, three, four, and five—while keeping everything else the same.

# Elbow Method

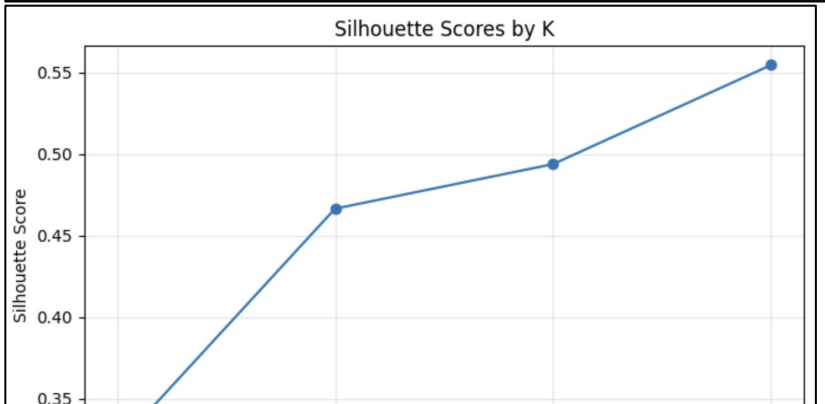



| | K | Inertia | Silhouette Score | Clusters Found |
|---|---|---|---|---|
| 0 | 2 | 269.6910 | 0.3213 | 2 |
| 1 | 3 | 157.7040 | 0.4666 | 3 |
| 2 | 4 | 108.9213 | 0.4939 | 4 |
| 3 | 5 | 65.5684 | 0.5547 | 5 |

The elbow method compares the inertia for different numbers of clusters. Inertia measures how far the customers are from their assigned cluster centroids, so lower values generally indicate tighter clusters. The largest decrease in my results occurred between two and three clusters. This makes three a reasonable simpler candidate, although the elbow is not completely clear.

# Silhouette Scores

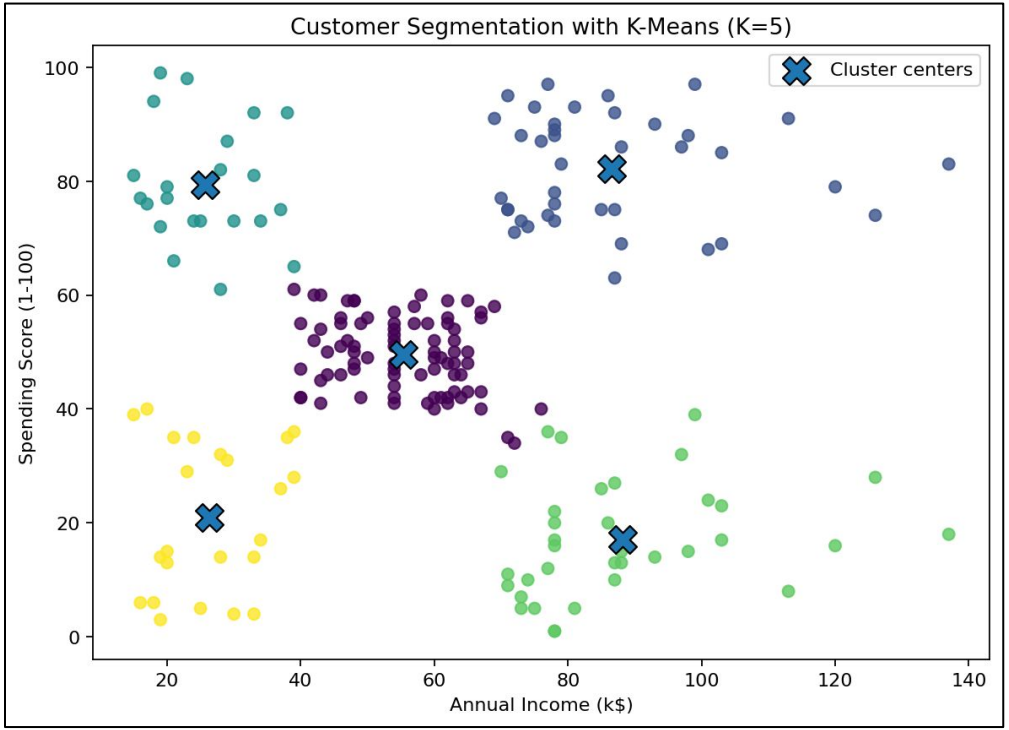
| K | Silhouette score |
|---|---|
| 2 | 0.3213 |
| 3 | 0.4666 |
| 4 | 0.4939 |
| 5 | 0.5547 |




- Higher scores generally indicate better cluster bond and separation.
- K=5 achieved the highest score among the four tested values with 0.5547.
- The elbow method and silhouette score do not identify one clear winner.

**Important Result:** The silhouette score favors five clusters among those tested, while the elbow method suggests that three clusters may be a simpler choice. Neither metric should be interpreted in isolation.

# 5 Clusters Visualizations




Key:

- X-Axis: Annual Income
- Y-axis: Spending Score
- Each Color: A different cluster
- Each Point: One customer

The graph helps us see patterns, but the colors themselves are simply cluster labels—not rankings or categories supplied in advance.

# Interpreting Customer Groups

K=5 Results:

| Cluster | Customers | Avg. income | Avg. spending |
|---|---|---|---|
| 0 | 81 | $55.30k | 49.52 |
| 1 | 39 | $86.54k | 82.13 |
| 2 | 22 | $25.73k | 79.36 |
| 3 | 35 | $88.20k | 17.11 |
| 4 | 23 | $26.30k | 20.91 |

**Interpretation:**

- Cluster 1: higher income and higher spending.
- Cluster 3: higher income and lower spending.
- Cluster 2: lower income and higher spending.
- Cluster 4: lower income and lower spending.
- Cluster 0: middle-range average income and spending.

By comparing Clusters 1 and 3, they both have high average incomes, but Cluster 1 has a high spending score, while Cluster 3 has a much lower one. This suggests that income alone does not explain spending behavior. These groups could help a business explore different customer strategies, although further information would be needed to understand why customers behave this way.

# Limitations

**Limitations**

- Only two customer features were used.
- The dataset does not provide ground-truth cluster labels.
- Results can vary with initialization and feature selection.
- Only four values of Kwere tested.
- The best numerical score does not necessarily mean the most useful business segmentation.

There are several limitations to this experiment. I used only two features, tested only four cluster counts, and did not have known customer-group labels to verify the clusters against. Adding features such as purchase frequency or customer tenure could provide a more complete picture. I would also compare K-Means with other clustering methods.

# Key Takeaways & Future Work

- K-Means can reveal useful patterns in customer data.
- The choice of K affects the resulting customer groups.
- In this experiment, the silhouette score favored K=5, while the elbow method was less conclusive.
- Income alone does not explain spending behavior.
- Future Work: test additional features, examine more values of K, and compare clustering algorithms.

This experiment helped me understand that choosing the number of clusters is not always straightforward. I learned that different evaluation methods can suggest different results and that more clusters do not automatically mean better business insights. If I repeated the experiment, I would test additional K values, include more customer features, and compare K-Means with other clustering methods.

# References


- Ikotun, A. M., Ezugwu, A. E., Abualigah, L., Abuhaija, B., & Heming, J. (2023). K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data. *Information Sciences, 622*, 178–210. https://www.sciencedirect.com/science/article/abs/pii/S0020025522014633

- Kaggle. Mall Customer Segmentation Dataset

- Liu, Y., & Huang, S. (2025). *A survey on K-means clustering algorithms: Theoretical analysis and performance comparison*. arXiv. https://knightnemo.github.io/assets/files/clustering.pdf